

# **Full Stack Development with MERN**

## **Project Documentation**

### **1. Introduction**

**Project Title:** ShopEZ – Full Stack E-Commerce Platform

### **2. Project Overview**

**Purpose:** ShopEZ is a full-stack e-commerce web application built using the MERN stack, which includes MongoDB, Express.js, React.js, and Node.js. The platform is designed to provide users with a smooth online shopping experience where they can browse products, add items to their cart, place orders, and track their purchases.

In addition to customer-facing features, ShopEZ also includes an admin dashboard that allows administrators to manage products, orders, users, inventory, and promotional banners efficiently.

The main objective of this project is to apply and demonstrate full-stack web development concepts in a real-world scenario. It showcases how modern technologies can be combined to create a scalable, secure, and user-friendly e-commerce platform.

### **Features**

#### **User Features**

- User Registration and Login using JWT Authentication
- Product Browsing and Search
- Category-Based Product Filtering
- Product Reviews and Ratings
- Shopping Cart Management
- Order Placement and Checkout
- Order Tracking System
- Order Cancellation
- User Profile Management
- Order History

#### **Admin Features**

- Secure Admin Authentication
- Dashboard Analytics

- Product Management
- Order Management
- User Management
- Banner Management
- Inventory Management

### **3. Architecture**

#### **Frontend Architecture (React.js)**

The frontend is developed using React.js and Vite. React Router DOM handles navigation between pages. Context API is used for global state management and Axios is used for communication with backend APIs.

Frontend Components:

- Home Page
- Login/Register
- Product Listing
- Product Details
- Cart
- Checkout
- Profile
- Admin Dashboard

#### **Backend Architecture (Node.js & Express.js)**

The backend follows a RESTful architecture built with Node.js and Express.js.

Main Components:

- Controllers
- Routes
- Middleware
- Models
- Authentication Services

Responsibilities:

- Handle API requests
- Authenticate users/admins

- Manage products
- Process orders
- Maintain cart data

### **Database Architecture (MongoDB)**

MongoDB is used as the NoSQL database. Mongoose provides schema modeling and database interaction.

Collections:

- Users
- Products
- Orders
- Carts
- Admins

Database Relationships:

- User → Orders
- User → Cart Items
- Product → Reviews
- Admin → Banners

## **4. Setup Instructions**

• **Prerequisites:** Install the following software:

- Node.js (v16 or above)
- MongoDB
- Git
- VS Code

### **Installation**

#### **Clone Repository**

```
git clone https://github.com/YOURUSERNAME/ShopEZ.git
```

```
cd ShopEZ
```

#### **Backend Setup**

```
cd server
```

```
npm install
```

Create a .env file:

MONGO\_URI=mongodb://localhost:27017/shopez

PORT=8000

JWT\_SECRET=shopez\_secret\_key\_2024

Start backend server:

npx nodemon index.js

## Frontend Setup

cd client

npm install

npm run dev

## Start MongoDB

# 5. Folder Structure

## Client Structure

client/

|

├── src/

| ├── api/

| ├── components/

| ├── context/

| ├── pages/

| └── App.jsx

| └── main.jsx

|

└── package.json

## Server Structure

server/

|

├── controllers/

├── models/

└── routes/

```
|— middleware/  
|— index.js  
└— package.json
```

## 6. Running the Application

### Frontend

cd client

npm run dev

Frontend URL:

<http://localhost:5173>

### Backend

cd server

npm start

Backend URL:

<http://localhost:8000>

### Admin Panel

<http://localhost:5173/admin>

**Backend:** npm start in the server directory.

## 7. API Documentation

### User APIs

#### Register User

POST /api/users/register

Request:

```
{  
  "username": "Rahul",  
  "email": "rahul@gmail.com",  
  "password": "123456"  
}
```

Response:

```
{  
  "message": "User Registered Successfully"
```

```
}
```

## Login User

POST /api/users/login

Response:

```
{  
  "token": "jwt_token"  
}
```

## Product APIs

| Method | Endpoint          | Description        |
|--------|-------------------|--------------------|
| GET    | /api/products     | Get all products   |
| GET    | /api/products/    | Get single product |
| POST   | /api/products/add | Add product        |
| DELETE | /api/products/    | Delete product     |

## Cart APIs

| Method | Endpoint      |
|--------|---------------|
| GET    | /api/cart     |
| POST   | /api/cart/add |
| DELETE | /api/cart/    |

## Order APIs

| Method | Endpoint             |
|--------|----------------------|
| POST   | /api/orders/place    |
| GET    | /api/orders/myorders |
| PUT    | /api/orders/cancel/  |

## Admin APIs

| Method | Endpoint          |
|--------|-------------------|
| POST   | /api/admin/login  |
| GET    | /api/admin/orders |

GET    /api/admin/users

PUT    /api/admin/orders/

## 8. Authentication

Authentication is implemented using JSON Web Tokens (JWT).

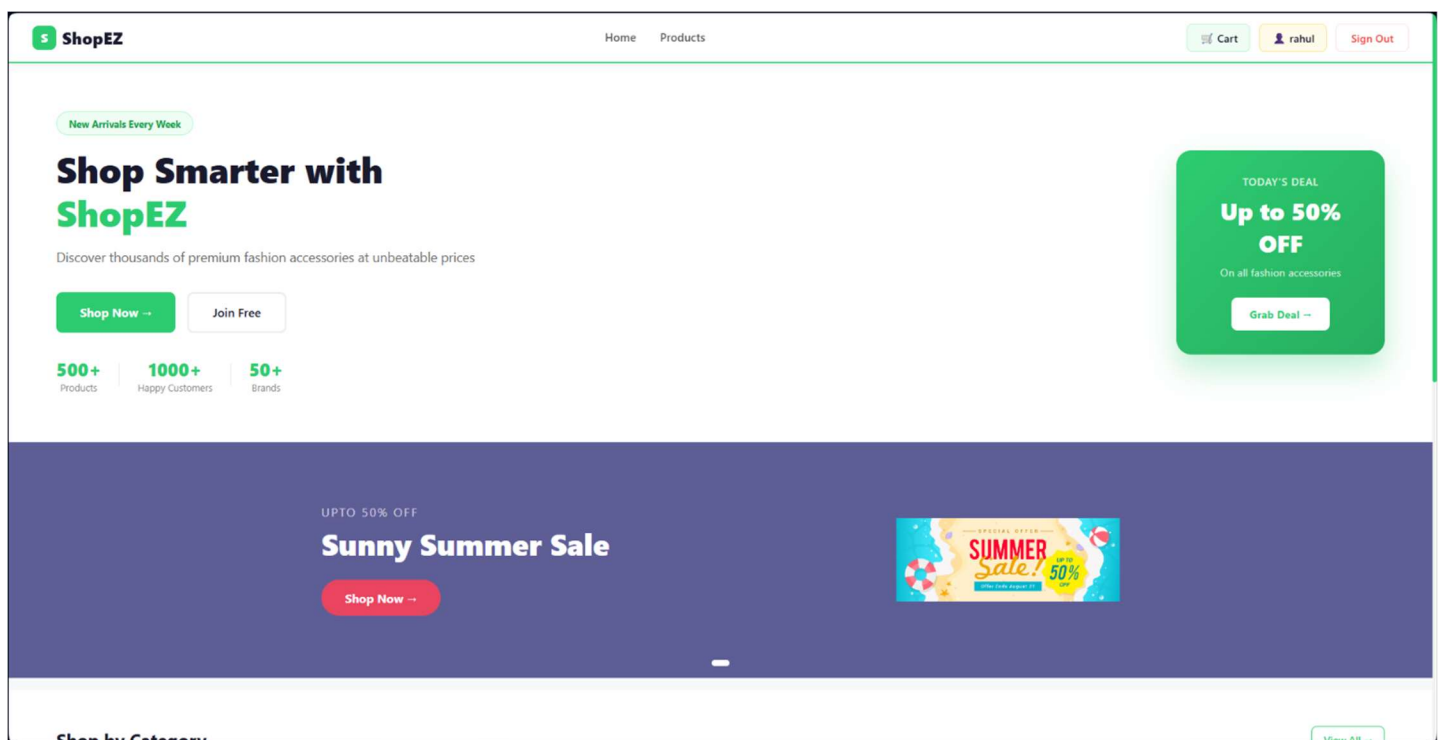
Process:

1. User/Admin enters credentials.
2. Server validates credentials.
3. JWT token is generated.
4. Token is stored in local Storage.
5. Protected routes verify token before granting access.

Security Measures:

- Password Hashing using bcryptjs
- JWT-based Authentication
- Protected API Routes
- Role-Based Authorization

## 9. User Interface



ShopEZ

HomeProducts

CartrahulSign Out

Shop by Category

View All →

Bracelets

Necklaces

Earrings

Rings

Handbags

Accessories

Why Choose ShopEZ?

Easy Discovery

Browse hundreds of products with powerful search and filters

Secure Checkout

Your payments and personal data are always protected

Fast Delivery

Get your orders delivered at lightning speed

Premium Quality

Only the best products from verified sellers

Ready to Start Shopping?

Join thousands of happy customers today

Create Free Account →

ShopEZ

HomeProducts

CartrahulSign Out

All Products

Bracelets

Necklaces

Earrings

Rings

Handbags

Accessories

Beauty

Electronics

Sports

Clothing

Footwear

Watches

Sunglasses

8% off



ShopEZ

Adidas FIFA World Cup 26 League Football

★★★★★ (0)

₹14719 ₹15999

20% off

39.62cm, Windows 11

ShopEZ

Dell G Series, 13th Gen Intel Core i5-13450HX, NVIDIA RTX 3050-6GB, 16GB DDR5, 512GB SSD, FHD, 15.6"/>39.62cm, Windows 11

★★★★★ (0)

₹84799 ₹105999

8% off

ShopEZ

Apple 2026 MacBook Air 13" Laptop with M5 chip: AI and Apple Intelligence, 34.46 cm (13.6") Liquid Retina Display, 16GB Unified Memory, 512GB SSD Storage, 12MP Center Stage Camera, Touch ID

★★★★★ (0)

₹110308 ₹119900

23% off

ShopEZ

Apple iPhone 17 Pro Max 1 TB: 17.42 cm (6.9") Display with Promotion, A19 Pro Chip, Best Battery Life in Any iPhone Ever, Pro Fusion Camera System, Center Stage Front Camera

★★★★★ (1)

₹148499 ₹149999

10% off



ShopEZ

Seiko Analog Orange Dial Men's Watch-SSK005K1

★★★★★ (0)

₹40500 ₹45000



ShopEZ

Ray-Ban Meta (Gen 2), Wayfarer, Matte Black | Smart AI Glasses for Men, Women — 2X Battery Life — 3K Ultra HD Resolution — 12 MP Ultra-Wide Camera, Audio, Video — Transitions® Grey Lenses

★★★★★ (0)

₹44999



ShopEZ

Cetaphil Gentle Skin Hydrating Face Wash 236ml, Paraben Free, Sulphate-Free Gentle Skin Hydrating Cleanser with Niacinamide, Vitamin B5 for Dry to Normal, Sensitive Skin

★★★★★ (0)

₹917



ShopEZ

adidas ADIZERO EVO SL CLOUD WHITE/CORE BLACK/CLOUD WHITE

★★★★★ (0)

₹16319

Out of Stock



ShopEZ

HomeProducts

CartrahulSign Out

Shopping Cart

2 items in your cart

Continue Shopping



Seiko Analog Orange Dial Men's Watch-SSK005K1

₹45000 per item

Subtotal: ₹45,000

-1+

Remove



Ray-Ban Meta (Gen 2), Wayfarer, Matte Black | Smart AI Glasses for Men, Women — 2X Battery Life — 3K Ultra HD Resolution — 12 MP Ultra-Wide Camera, Audio, Video — Transitions® Grey Lenses

₹44999 per item

Subtotal: ₹44,999

-1+

Remove

Order Summary

Price (2 items)₹89,999

Delivery ChargesFREE

Total Amount₹89,999

You save ₹0 on this order!

Proceed to Checkout

Safe and Secure Payments

We Accept all Payment Methods:



ShopEZ

HomeProducts

CartrahulSign Out

R

Hello, rahul!

Email: rahul@gmail.com

Member since 2026

Sign Out

1Total Orders

0Delivered

0Pending

₹149,999Total Spent

My Account

My Orders

Account Details

Saved Addresses

Payment Methods

Continue Shopping

My Orders

ORDER PLACED17 June 2026

TOTAL₹149,999

PAYMENTCash on Delivery

ORDER ID#68C6325D

Cancelled

Apple iPhone 17 Pro Max 1 TB: 17.42 cm (6.9") Display with Promotion, A19 Pro Chip, Best Battery Life in Any iPhone Ever, Pro Fusion Camera System, Center Stage Front Camera;

Quantity: 1

Price: ₹149999 per item

Rahul Srimaan, 8966936450, 432 A Street No1,Green Park,AS Rao Nagar ECIL, Hyderabad, Telangana - 500040

Order Cancelled

This order has been cancelled

ShopEZ

HomeProducts

Cart

rahul

Sign Out

Checkout

1 Delivery Address

FULL NAME \*

Enter your full name

PHONE NUMBER \*

10-digit mobile number

PINCODE \*

6-digit pincode

CITY \*

Enter city

STREET ADDRESS \*

House No., Building, Street, Area

LANDMARK (OPTIONAL)

e.g. Near City Mall

STATE \*

Select State

2 Payment Method

Credit Card

Debit Card

UPI Payment

Cash on Delivery

3 Order Summary

Seiko Analog Orange Dial Men's Watch-SSK005K1

Qty: 1

₹45,000

Ray-Ban Meta (Gen 2), Wayfarer, Matte Black | Smart AI Glasses for Men, Women — 2X Battery Life — 3K Ultra HD Resolution — 12 MP Ultra-Wide Camera, Audio, Video — Transitions® Grey Lenses

Qty: 1

₹44,999

Price (2 items)

₹89,999

Delivery

FREE

Total Amount

₹89,999

Place Order →

100% Secure Payments

ShopEZ

ADMIN PORTAL

Admin

Administrator

MAIN MENU

Dashboard

Orders

Products

Add Product

Banners

Users

Sign Out

Dashboard

Thu Jun 18 2026

TOTAL REVENUE

₹470,783

13 orders

TOTAL ORDERS

13

0 pending

PRODUCTS

9

In catalog

CUSTOMERS

7

Registered

DELIVERED

6

Completed

PENDING

0

To process

Recent Orders

| PRODUCT         | PRICE | QTY | PAYMENT          | STATUS    | DATE      |
|-----------------|-------|-----|------------------|-----------|-----------|
| Gold Bangle     | ₹1499 | 1   | Credit Card      | Delivered | 6/13/2026 |
| Gold Bangle     | ₹1499 | 3   | Cash on Delivery | Delivered | 6/14/2026 |
| Gold Bangle     | ₹1499 | 1   | UPI              | Delivered | 6/14/2026 |
| Gold Bangle     | ₹1499 | 4   | Cash on Delivery | Delivered | 6/14/2026 |
| Silver Necklace | ₹2499 | 1   | Cash on Delivery | Shipped   | 6/14/2026 |

S

ShopEZ

ADMIN PORTAL

A

Admin

Administrator

MAIN MENU

Dashboard

Orders

Products

Add Product

Banners

Users

Sign Out

Orders

Thu Jun 18 2026

A Admin

All Orders (13)

| PRODUCT                                                                                                                                                                       | PRICE   | QTY | ADDRESS         | PAYMENT          | STATUS     | UPDATE     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-----|-----------------|------------------|------------|------------|
| Gold Bangle                                                                                                                                                                   | ₹1499   | 1   | 123 Main St...  | Credit Card      | Delivered  | Delivered  |
| Gold Bangle                                                                                                                                                                   | ₹1499   | 3   | park lane,Hy... | Cash on Delivery | Delivered  | Delivered  |
| Gold Bangle                                                                                                                                                                   | ₹1499   | 1   | anurag univ...  | UPI              | Delivered  | Delivered  |
| Gold Bangle                                                                                                                                                                   | ₹1499   | 4   | anurag univ...  | Cash on Delivery | Delivered  | Delivered  |
| Silver Necklace                                                                                                                                                               | ₹2499   | 1   | anurag univ...  | Cash on Delivery | Shipped    | Shipped    |
| Pearl Earrings                                                                                                                                                                | ₹1299   | 1   | Hyderabad       | UPI              | Shipped    | Shipped    |
| Silver Necklace                                                                                                                                                               | ₹2499   | 1   | Flat No. 402... | Credit Card      | Processing | Processing |
| adidas ADIZERO EVO SL CLOUD WHITE/CORE BLACK/CLOUD WHITE                                                                                                                      | ₹15999  | 1   | Tech Vantag...  | Cash on Delivery | Delivered  | Delivered  |
| Leather Handbag                                                                                                                                                               | ₹3999   | 1   | Plot 565, Ro... | Credit Card      | Processing | Processing |
| Seiko Analog Orange Dial Men's Watch-SSK005K1                                                                                                                                 | ₹45000  | 1   | Rohan Redd...   | UPI              | Cancelled  | Pending    |
| Dell G Series, 13th Gen Intel Core i5-13450HX, NVIDIA RTX 3050-6GB, 16GB DDR5, 512GB SSD, FHD, 15.6"/39.62cm, Windows 11                                                      | ₹105999 | 1   | Rohan Redd...   | Cash on Delivery | Cancelled  | Pending    |
| Samsung Galaxy S25 Ultra 5G AI Smartphone (Titanium Whitesilver, 12GB RAM, 256GB Storage), 200MP Camera, S Pen Included, Long Battery Life                                    | ₹129999 | 1   | Rohan Redd...   | Cash on Delivery | Delivered  | Delivered  |
| Apple iPhone 17 Pro Max 1 TB: 17.42 cm (6.9") Display with Promotion, A19 Pro Chip, Best Battery Life in Any iPhone Ever, Pro Fusion Camera System, Center Stage Front Camera | ₹149999 | 1   | Rahul Srima...  | Cash on Delivery | Cancelled  | Pending    |

S

ShopEZ

ADMIN PORTAL

A

Admin

Administrator

MAIN MENU

Dashboard

Orders

Products

Add Product

Banners

Users

Sign Out

Products

Thu Jun 18 2026

A Admin

All Products (9)

| IMAGE                                                                                                                                                                                                                                                                                                                                 | NAME                                                                                                                                                                                          | CATEGORY    | PRICE   | STOCK | DISCOUNT | ACTION |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------|-------|----------|--------|
|                                                                                                                                                                                                                                                    | Adidas FIFA World Cup 26 League Football                                                                                                                                                      | Sports      | ₹15999  | 100   | 8%       | Delete |
|                                                                                                                                                                                                                      | Dell G Series, 13th Gen Intel Core i5-13450HX, NVIDIA RTX 3050-6GB, 16GB DDR5, 512GB SSD, FHD, 15.6"/39.62cm, Windows 11                                                                      | Electronics | ₹105999 | 120   | 20%      | Delete |
|  | Apple 2026 MacBook Air 13" Laptop with M5 chip: AI and Apple Intelligence, 34.46 cm (13.6") Liquid Retina Display, 16GB Unified Memory, 512GB SSD Storage, 12MP Center Stage Camera, Touch ID | Electronics | ₹119900 | 90    | 8%       | Delete |
|                                                                                                                                                                                                                                                    | Samsung Galaxy S25 Ultra 5G AI Smartphone (Titanium Whitesilver, 12GB RAM, 256GB Storage), 200MP Camera, S Pen Included, Long Battery Life                                                    | Electronics | ₹129999 | 119   | 23%      | Delete |
|                                                      | Apple iPhone 17 Pro Max 1 TB: 17.42 cm (6.9") Display with Promotion, A19 Pro Chip, Best Battery Life in Any iPhone Ever, Pro Fusion Camera System, Center Stage Front Camera                 | Electronics | ₹149999 | 25    | 1%       | Delete |
|                                                                                                                                                                                                                                                    | Seiko Analog Orange Dial Men's Watch-SSK005K1                                                                                                                                                 | Watches     | ₹45000  | 45    | 10%      | Delete |
|                                                                                                                                                                                                                                                    | Ray-Ban Meta (Gen 2), Wayfarer, Matte Black   Smart AI Glasses for Men, Women — 2X Battery Life — 3K Ultra HD Resolution — 12 MP Ultra-Wide Camera, Audio, Video — Transitions® Grey Lenses   | Sunglasses  | ₹44999  | 90    | 0%       | Delete |
|                                                                                                                                                                                                                                                    | Cetaphil Gentle Skin Hydrating Face Wash 236ml, Paraben Free, Sulphate-Free Gentle Skin Hydrating Cleanser with Niacinamide, Vitamin B5 for Dry to Normal, Sensitive Skin                     | Beauty      | ₹899    | 250   | -2%      | Delete |
|                                                                                                                                                                                                                                                    | adidas ADIZERO EVO SL CLOUD WHITE/CORE BLACK/CLOUD WHITE                                                                                                                                      | Footwear    | ₹15999  | 9     | -2%      | Delete |

ShopEZADMIN PORTAL

AdminAdministrator

MAIN MENU

Dashboard

Orders

Products

Add Product

Banners

Users

Sign Out

Add ProductThu Jun 18 2026

Add New Product

PRODUCT NAME

e.g. Gold Bangle

PRICE (₹)

e.g. 1499

CATEGORY

e.g. bracelets

BRAND

e.g. Tanishq

STOCK

e.g. 50

DISCOUNT (%)

e.g. 10

MAIN IMAGE URL

https://images.unsplash.com/...

ADDITIONAL IMAGES (ONE URL PER LINE)

https://image2.jpg  
https://image3.jpg

HIGHLIGHTS (ONE PER LINE)

100% Pure Gold  
Free Shipping  
Easy Returns

SPECIFICATIONS

MATERIAL

e.g. 18K Gold

WEIGHT

e.g. 10g

DIMENSIONS

e.g. 6cm x 2cm

COLOR

e.g. Gold

ShopEZ

HomeProducts

LoginRegister

Search for products, brands and more...

SearchSort By

Categories

All Products

Bracelets

Necklaces

Earrings

Rings

Handbags

Accessories

Beauty

Electronics

Sports

Clothing

Footwear

Watches

Sunglasses

Electronics

4 products found

20% off



ShopEZ

Dell G Series, 13th Gen Intel Core i5-13450HX, NVIDIA RTX 3050-6GB, 16GB DDR5, 512GB SSD, FHD, 15.6"/39.62cm, Windows 11

★★★★★ (3)

₹84799 ₹105999

8% off



ShopEZ

Apple 2026 MacBook Air 13" Laptop with M5 chip: AI and Apple Intelligence, 34.46 cm (13.6") Liquid Retina Display, 16GB Unified Memory, 512GB SSD Storage, 12MP Center Stage Camera, Touch ID:

★★★★★ (2)

₹110308 ₹119999

23% off



ShopEZ

Samsung Galaxy S25 Ultra 5G AI Smartphone (Titanium Whitesilver, 12GB RAM, 256GB Storage), 200MP Camera, S Pen Included, Long Battery Life

★★★★★ (1)

₹100099 ₹129999

1% off



ShopEZ

Apple iPhone 17 Pro Max 1TB: 17.42 cm (6.9") Display with Promotion, A19 Pro Chip, Best Battery Life in Any iPhone Ever, Pro Fusion Camera System, Center Stage Front Camera:

★★★★★ (1)

₹148499 ₹149999

## 10. Testing

### Testing Strategy

The application was tested manually using browser testing and Postman API testing.

### Testing Tools

- Postman
- Chrome Developer Tools
- MongoDB Compass

### Sample Test Cases

| Test Case    | Expected Result           |
|--------------|---------------------------|
| User Login   | Successful Authentication |
| Add Product  | Product Added             |
| Add to Cart  | Cart Updated              |
| Place Order  | Order Created             |
| Cancel Order | Status Updated            |
| Admin Login  | Dashboard Access          |

## 11. Screenshots or Demo

### Demo Video

GitHub Repository: <https://github.com/rahulsrimaan/VIP-C2-ShopEZ>

## 12. Known Issues

- Product images currently use URLs instead of file uploads.
- No online payment gateway integration.
- Limited analytics for administrators.
- Search functionality can be optimized further.
- No wish list feature.

## 13. Future Enhancements

- Razor pay Payment Gateway Integration
- Stripe Payment Integration
- Cloudinary Image Uploads
- Email Notifications

- Wishlist Functionality
- Product Comparison
- Coupon and Discount Management
- Advanced Analytics Dashboard
- Mobile Application using React Native
- AI-Based Product Recommendations